SE-24 Group

TEAM PROJECT BRIEF & TECHNICAL SPECIFICATION

Complete Subsystem Breakdown, Frozen Pinout & Implementation Roadmap

TEAM PROJECT BRIEF & TECHNICAL SPECIFICATION

Project: Smart Control and Monitoring System for CoCIS (Makerere University)

Document Owner: Kayeera Nathan (Group Leader & System Architect)

Group: SE-24 (BSE III - Academic Year 2026/2027)

Reference Document: Updated from SE-24 Project Kickoff CoCIS Smart Control and Monitoring System.docx the link I shared from the drive

Tooling: PlatformIO in VS Code + Wokwi Simulation (framework = espidf)

Communication Bus: Wireless ESP-NOW (2.4 GHz Peer-to-Peer Wi-Fi)

Standardized Node Names: ENTERANCE_MASTER, ROOM_NODE_1, ROOM_NODE_2

Coursework Submission Deadline: Monday, 2 November 2026 on MUELE

In-Person Defense : Monday, 2 November 2026 (Attendance Compulsory)

1. Executive Summary & Key Architectural Updates

Hi Team (David, Mark, Jovan, Laurinda),

Following our initial project kickoff (SE-24 Project Kickoff CoCIS Smart Control and Monitoring System.docx), this comprehensive specification unifies our system architecture, hardware pin mappings, communication protocols, and individual member workloads.

Major Upgrades from the Initial Kickoff:

1. Wireless ESP-NOW Communication (No Cables across Rooms): Instead of running long, fragile physical UART serialcables between the entrance and the two lecture rooms, we have upgraded to ESP-NOW (2.4 GHz connectionless Wi-Fi). This operates without a router, provides sub-5ms latency, includes built-in hardware MAC-layer delivery acknowledgments, and directly scores bonus innovation marks with the lecturer.
2. Standardized Node Naming Convention: All firmware modules, PlatformIO environments, and Wokwi simulation chipsare standardized as:
   - ENTERANCE_MASTER (Entrance Terminal, SD Card, CLI Console, FreeRTOS Gateway - ESP32 #1)
   - ROOM_NODE_1 (Lecture Room 1 Controller - ESP32 #2)
   - ROOM_NODE_2 (Lecture Room 2 Controller - ESP32 #3)
3. PlatformIO & VS Code Workspace: The full codebase has been pre-generated as a monorepo in **SE-24-CoCIS-Smart-System/**. You can build and simulate your own node individually, or run the entire 3-node system simultaneously in Wokwi!

2. Group Members, Roles & Technical Deliverables

No	Member Name	Registration No	Role	Core Technical Deliverables
1	Kayeera Nathan	24/U/25913/EVE	Group Leader, Architect & Integrator	System architecture, FreeRTOS mutexes/queues concurrency layer, ESP-NOW protocol layer, Emergency Evacuation mode, Heartbeat monitoring, master integration, final report compilation & MUELE submission.
2	Kasozi David Paul	24/U/25905/EVE	Entrance Terminal Developer	Dual RFID reader SPI drivers (Entry & Exit), Servo PWM actuators with 5s timed relock, Access feedback LEDs, Anti-passback algorithm, Intrusion alarm buzzer & lockout.

3	Zziwa Mark Alex	24/U/11978/EVE	Lecture Room Terminals Developer	PIR motion sensing with software debounce, 30s no-motion countdown timer, Ambient light LDR thresholding (Daylight-aware logic), Light relay actuator, 2s ESP-NOW telemetry dispatch.
4	Sseku Jovan	24/U/11172/EVE	Storage & Data Management Developer	SPI MicroSD card module setup with FATFS, CSV file managers (users.csv, access_log.csv, occupancy.csv, config.txt), Timestamp helper, Crash recovery from unexpected reboot.
5	Namuzibwa Laurinda	24/U/1472	Interface, Security & QA Lead	Serial CLI parser and navigation menus over USB-UART0, SHA-256 administrative password authentication, session timeouts, schedule configuration commands, test plan execution & bug log.

3. Complete Hardware Pin Allocations (FROZEN SPECIFICATION)

Please wire your simulation components in Wokwi strictly according to these pin mappings. Do not change pins without notifying the whole group in chat.

3.1 Node 1: ‘ENTERANCE_MASTER‘ (ESP32 #1 - MAC: ‘24:0A:C4:00:01:00‘)

Peripheral / Signal	ESP32 GPIO	Direction / Type	Detailed Functional Description
RFID Entry Reader CS	GPIO 21	Output (SPI CS)	Dedicated Chip Select for Inbound RC522 reader.
RFID Exit Reader CS	GPIO 5	Output (SPI CS)	Dedicated Chip Select for Outbound RC522 reader.
Shared RFID SPI MOSI	GPIO 23	Output (SPI MOSI)	Master Out Slave In for dual RC522 readers.
Shared RFID SPI MISO	GPIO 19	Input (SPI MISO)	Master In Slave Out for dual RC522 readers.
Shared RFID SPI SCK	GPIO 18	Output (SPI CLK)	Serial Clock line for dual RC522 readers.
RFID Reset (RST)	GPIO 22	Output (Reset)	Shared hardware reset line for RFID modules.
Entry Door Actuator	GPIO 13	Output (PWM)	Servo Motor: 0 closed, 90 open for 5 seconds on valid scan.
Exit Door Actuator	GPIO 15	Output (PWM)	Servo Motor: 0 closed, 90 open for 5 seconds on valid exit.
Access Granted LED	GPIO 2	Output (Active High)	Green LED: Flashes on authorized entry or exit.
Access Denied LED	GPIO 4	Output (Active High)	Red LED: Flashes on rejected scan or curfew attempt.
Alarm / Lockout Buzzer	GPIO 12	Output (Active High)	Buzzer: Pulses on 3 invalid attempts or unauthorized exit.
Emergency Pushbutton	GPIO 32	Input (Pull-up)	Direct hardware interrupt: forces all doors open & evac alert.
MicroSD Card CS	GPIO 27	Output (SPI CS)	Dedicated Chip Select for MicroSD Card module.
MicroSD SPI MOSI	GPIO 23	Output (Shared SPI)	Data line to SD Card.

MicroSD SPI MISO	GPIO 19	Input (Shared SPI)	Data line from SD Card.
MicroSD SPI CLK	GPIO 18	Output (Shared SPI)	Clock line to SD Card.
Serial CLI Terminal	TX0 / RX0	UART (GPIO 1 & 3)	115200 Baud console for admin login and monitoring.
Wireless Transceiver	Internal RF	2.4 GHz RF Antenna	Wireless ESP-NOW gateway transceiver.

3.2 Node 2 & Node 3: ‘ROOM_NODE_1‘ & ‘ROOM_NODE_2‘ (ESP32 #2 & #3)

Peripheral / Signal	ESP32 GPIO	Direction / Type	Detailed Functional Description
PIR Motion Sensor	GPIO 14	Input (Interrupt)	Detects human presence; resets the 30-second countdown timer.
Ambient Light Sensor	GPIO 34	Analog Input (ADC1)	LDR photocell voltage divider; measures ambient light (lux).
Lighting Control Relay	GPIO 12	Output (Active High)	Controls room lights: Closed (ON) / Open (OFF).
Heartbeat Status LED	GPIO 2	Output (Active High)	Flashes briefly every 2 seconds upon wireless telemetry transmit.
Wireless Transceiver	Internal RF	2.4 GHz RF Antenna	Wireless ESP-NOW transmitter (Room 1 MAC: 24:0A:C4:00:02:00).

4. Wireless Communication Protocol (ESP-NOW)

All inter-microcontroller data exchanges occur wirelessly over the 2.4 GHz radio space using standardized packet frames defined in components/common_protocol/include/protocol.h: in the git repo I am soon sharing in the group chat

Key Protocol Rules:

1. Periodic Telemetry: ROOM_NODE_1 and ROOM_NODE_2 broadcast their state every 2 seconds (MSG_TYPE_ROOM_STATUS):
   occupied (1 = Yes, 0 = No) lights_on (1 = ON, 0 =
   OFF)
   light_level_lux (Ambient illumination in lux)
   seconds_active (Total seconds lights were turned off while room was vacant)
2. Delivery Confirmation: Unicast packets trigger automatic 802.11 MAC-layer ACKs. Senders checkesp_now_register_send_cb and retry up to 3 times on transmission failure.
3. Emergency Broadcast: ENTERANCE_MASTER broadcasts MSG_TYPE_EMERGENCY_ALERT to MAC
   FF:FF:FF:FF:FF:FF. Both room nodes immediately force their light relays ON for occupant evacuation.
4. Heartbeat & Offline Detection: If ENTERANCE_MASTER receives no telemetry from a room node for more than 10 seconds, it flags an OFFLINE alert on the administrator dashboard.

5. Storage Schemas & Database Structure (SD Card)

Sseku Jovan manages the MicroSD card (SPI/FATFS). Only ENTERANCE_MASTER writes to disk.

1. /sdcard/users.csv (Authorized Personnel Database):

```
tag_id,name,role,is_active    TAG_1001,Kayeera Nathan,admin,1
TAG_1002,Kasozi David,staff,1
TAG_1003,Zziwa Mark,student,1
TAG_1004,Sseku Jovan,student,1
TAG_1005,Namuzibwa Laurinda,staff,1
```

2. /sdcard/access_log.csv (Permanent Security Audit Trail):

```
timestamp,tag_id,user_name,direction,decision,reason
2026-10-25 08:15:20,TAG_1001,Kayeera Nathan,ENTRY,GRANTED,Authorized
2026-10-25 21:05:11,TAG_1003,Zziwa Mark,ENTRY,DENIED,Outside Schedule
2026-10-25 21:10:02,TAG_1004,Sseku Jovan,EXIT,DENIED,Curfew Lockdown
2026-10-25 21:15:00,SYSTEM,Admin,EMERGENCY,ACTIVE,Evacuation Triggered
```

3. /sdcard/occupancy.csv (Dynamic Crash Recovery Mirror):

Updated on every valid entry and exit so building occupancy is restored if power drops.

4. /sdcard/config.txt (System Configuration):

```
schedule_start=07:00  schedule_end=20:00  lockout_attempts=3  lockout_duration_sec=60  no_motion_timeout_sec=30  heartbeat_interval_sec=2
```

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6. Serial CLI Commands & Administrator Interface

Namuzibwa Laurinda manages the terminal interface running on ENTERANCE_MASTER (USB-UART0 at 115200 Baud):

Command	Arguments	Required Role	Functionality
help	none	Public	Displays all available commands and syntax.
status	none	Public	Prints system health, SD card status, and wireless room links.
occupancy	none	Public	Lists the full names of everyone currently inside the building.
rooms	none	Public	Displays live occupancy, lighting, and lux levels for Rooms 1 & 2.
lights	none	Public	Displays current relay states and energy savings statistics.
login	<user> <pass>	Public	Authenticates admin session using SHA-256 password hash.
logout	none	Authenticated	Terminates active admin session.
schedule show	none	Public	Displays current entry/exit permitted hours.
schedule set	<start> <end>	Admin Only	Updates operating hours in RAM and writes changes to SD card.
logs	[count]	Admin Only	Displays the last N access log records.
adduser	<tag> <name> <role>	Admin Only	Adds a new RFID tag to users.csv.
deluser	<tag>	Admin Only	Revokes credentials for a tag in users.csv.
energy	none	Admin Only	Prints estimated kilowatt-hours saved by daylight and no-motion logic.
emergency	‘<on\ off>‘	Admin Only	

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7. Innovation Features Committed for High Marks

Section 6 of the coursework brief heavily rewards system innovation. Our committed features are:

1. Industrial 2.4 GHz ESP-NOW Wireless Mesh (Nathan): Replaces physical wires across lecture rooms with alow-latency, connectionless protocol with hardware MAC ACKs.
2. Strict After-Hours Curfew Lockdown (David & Nathan): Enforces Section 2.6: Individuals inside the building cannotleave outside authorized hours, preventing theft or after-hours breaches.
3. Anti-Passback Prevention (David & Jovan): Prevents card tailgating or sharing (a tag recorded as "Inside" cannot enteragain; an "Outside" tag cannot exit).

4. Daylight-Aware Smart Lighting (Mark): Prevents energy waste by inhibiting lights in occupied lecture rooms if ambientdaylight exceeds threshold (>250 lux).
5. Crash Recovery System (Jovan): Master restores occupant count and schedule settings directly from/sdcard/occupancy.csv upon reboot.
6. Intrusion Lockout Alarm (David & Laurinda): 3 consecutive invalid RFID taps trigger the buzzer and lock down theterminal for 60 seconds.
7. Emergency Evacuation Broadcast (Nathan & David): Single button or command triggers exit door unlock andbroadcasts light-on override to all rooms.

8. Step-by-Step Programme of Work (Key Dates)

Phase	Dates	Target Milestones	Primary Owners
Phase 0	Sat 24 - Sun 25 Oct	Architecture freeze, repo setup, Wokwi feasibility, role confirmations.	All (Nathan leads)
Phase 1	Sun 25 - Tue 27 Oct	Independent module development: RFID, Servos, PIR, SD FATFS, CLI.	David, Mark, Jovan, Laurinda
Phase 2	Wed 28 - Thu 29 Oct	Master node integration, FreeRTOS queue binding, schedule enforcement.	Nathan, David, Jovan
Phase 3	Thu 29 - Fri 30 Oct	Innovation hardening, concurrency testing, HARD CODE FREEZE (Fri 8:00 PM).	All
Phase 4	Fri 30 Oct - Sun 1 Nov	Report drafting, recording 3-5 min individual videos, mock presentation.	All (Laurinda formats, Nathan compiles)
Phase 5	Sun 1 - Mon 2 Nov	Final MUELE PDF submission (Sun Night) & Lecturer In-Person Defense.	All (Attendance compulsory)

9. Deliverables & Individual Video Requirements

Every member must produce:

1. Written Report Section: 1 to 2 pages on your assigned module + personal contribution paragraph.
2. Individual Video Demonstration (3 to 5 Minutes):
   - Screen recording showing your face / voice.
   - Explain your role and demonstrate the code sections you personally wrote.
   - Run a live Wokwi demo of your feature (e.g. David demos anti-passback, Mark demos PIR/LDR, Jovan demos SD crash recovery, Laurinda demos CLI login & schedule set, Nathan demos architecture, ESP-NOW and emergency evacuation).
   - Upload to YouTube (Unlisted) or Google Drive ("Anyone with link can view") and send the link in the group or to Nathan for overall compiling
3. In-Person Presentation: Monday, 2 November 2026. Only attending students receive marks!